

ARYENDRA PANDEY

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SKILLS

Languages: Python, C++, JavaScript, ARM Assembly

Web Tech: HTML, CSS, FastAPI, Next.js, TypeScript, Tailwind CSS

Cybersecurity: Wireshark, Splunk Cloud, Kali Linux, SQL

Electronics Tools: Multisim, LTSpice, Microwind, MATLAB, QualNet

WORK EXPERIENCE

Software Developer Intern

April 2026 – May 2026

Tridit Digital

- Worked on Adobe Experience Manager (AEM) development, customizing content components and templates for enterprise web solutions.
- Integrated AI solutions into AEM pipelines for intelligent content personalization and automated tagging.

IT Infrastructure Intern

June 2026 – Present

XtraNet Technologies Limited, Bhopal

- Working with the IT Infra Team on infrastructure operations, network management, and system administration.
- Gaining hands-on exposure to enterprise IT environments, client systems, and confidential infrastructure workflows.

SOFTWARE PROJECTS

Brain Tumor Detection System with Hardware-Aware Inference Optimization

GitHub

- Trained YOLOv8n on 7,200 MRI scans across 4 tumor classes achieving 89.18% mAP@0.5 with CUDA-accelerated inference at 6.23 ms/image on an RTX 3050 GPU.
- Implemented FP32 → INT8 static quantization achieving 47.89% model size reduction (5.94 MB → 3.09 MB) with only 0.48% mAP degradation for edge deployment feasibility.
- Designed an 8-bit fixed-point MAC unit and 3×3 convolution engine in Verilog HDL, correlating simulated hardware cycle counts with profiled model MACs to project theoretical accelerator latency.
- Deployed LayerCAM explainability with IoU scoring integrated into a Streamlit dashboard with real-time YOLO inference and FP32 vs INT8 benchmark panels.

QoS Degradation Predictor — Network ML System

GitHub

- Built a time-driven simulator modeling multi-flow packet-switched networks with FIFO bottleneck queues.
- Implemented a logistic regression predictor for real-time QoS degradation detection and metric analysis.
- Developed the core system in C++17 with a modular CMake-based architecture.

NeuroSentry — AI-Powered Cyber Threat Detection

GitHub

- Built a real-time threat detection pipeline converting system/audit logs into behavioral features and ML-based predictions.
- Implemented modular backend components for log parsing, feature extraction, REST APIs, and online inference.
- Developed a React + Vite dashboard to visualize live alerts, historical logs, and security statistics.

SoTrail — AI-Powered Travel Planning Platform

GitHub

- Developed an AI travel planner with real-time flight/hotel search using Amadeus API and multimodal transport options.
- Built a FastAPI backend with ChromaDB vector storage and a RAG chatbot powered by Llama models.
- Designed a Next.js + TypeScript frontend with chatbot panel, itinerary generator, and analytics.

EDUCATION

Manipal Institute of Technology, MAHE

2023–Present

B.Tech — Electronics and Communication Engineering

EXTRACURRICULAR ACTIVITIES

- **Management Committee Member** — Astronomy Club, Manipal
- **Organising Committee Member** — Revels 2025, Cultural Fest, Manipal
- **Core Committee Member** — TechTatva 2025, Technical Fest, Manipal
- **Management Committee Member** — Aaina Dramatics, Manipal

CERTIFICATIONS

AWS Cloud Practitioner

Google Cybersecurity

Meta Front-End Development

Meta JavaScript Programming

CEH – LearnKartS

UC Irvine: Business Writing